

Zero-Combustion AI Data Center

Magnetic-Assisted Power + Waterless Cooling

A pilot-ready plan for Texas data centers

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THE PROBLEM

448 TWh

Global DC energy last year • 208M tons CO₂ • 1.2T gallons water

945 TWh

Projected by 2030 — AI workloads more than double demand

Texas Grid

Peak strain, interconnection delays, and aquifer stress risk

CORE INSIGHT

No perpetual motion. Magnets don't create energy — they convert it with far less friction and loss. The real leverage point is *cooling*. Removing heat without water or chillers cuts 30–60% of a data center's parasitic load.

SOLUTION PILLARS

Power

Permanent-magnet direct-drive wind/solar + magnetic-bearing flywheels for ride-through. SMR or geothermal option for base load. Zero on-site combustion, zero diesel.

Cooling

Direct-to-chip two-phase dielectric (PG25) then full immersion. Closed loop, WUE ~0, PUE 1.05–1.1. 30–60% cooling energy cut vs. air.

Architecture

Designed for 50–100kW racks. Heat reuse for building HVAC and greenhouse.

Sensor-rich ops with real-time PUE/WUE telemetry.

PILOT PLAN

Phase 1

- 2 racks, 80–150kW total load

- Budget: \$250k (cooling skids, retrofit, sensors)

- Metrics: PUE, WUE, kWh saved, water gallons avoided

Phase 2

- Expand to 1MW pod in Carrollton

- Add flywheel + behind-meter solar

- Target: PUE ≤ 1.08 , WUE = 0, payback <3yr

Vendors are already shipping: **Vertiv CoolPhase Flex, ZutaCore HyperCool, Schneider Uniflair XCA**. Industry is moving from air to liquid at 50kW+ per rack. Texas SB-6 and water restrictions make zero-water designs a competitive advantage.



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5-Slide Pitch Outline

Zero-Combustion AI Data Center — Investor / Partner Deck

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Problem

- 448 TWh global DC use (2025) = 208M tCO₂, 1.2T gal water

- 945 TWh projected by 2030

- Texas: grid volatility, ERCOT peaks, water scarcity

- Result: higher opex, permitting risk, community pushback

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Physics Reality

- No "free energy" — magnets enable ultra-efficient conversion

- Permanent-magnet generators: >96% efficiency, no gearbox

- Magnetic bearings: near-zero friction for flywheels

- Cooling is 30–40% of load — that's the leverage

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Power Design

- Primary: grid + behind-meter PM wind/solar

- Ride-through: magnetic-bearing flywheels (not batteries)

- Base load option: SMR or geothermal — zero combustion

- Outcome: resilient, clean, < \$0.06/kWh blended

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Cooling Design

- Step 1: Direct-to-chip two-phase (PG25 dielectric)

- Step 2: Full immersion for GPU clusters

- Closed loop dry coolers — WUE ~0

- PUE 1.05–1.1, 30–60% cooling energy reduction, 50–100kW/rack ready

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Pilot & ROI

- Phase 1: 2 racks, \$250k, 6 months in Carrollton — measure PUE, WUE, kWh

- Phase 2: Scale to 1MW pod, add flywheel + solar

- ROI drivers: energy savings, water avoided, density premium, faster permitting

- Ask: site partner, \$250k pilot funding, vendor access — prove zero-water AI this year